

# Vivado 2019.1 - Example, HelloWorld

## Introduction

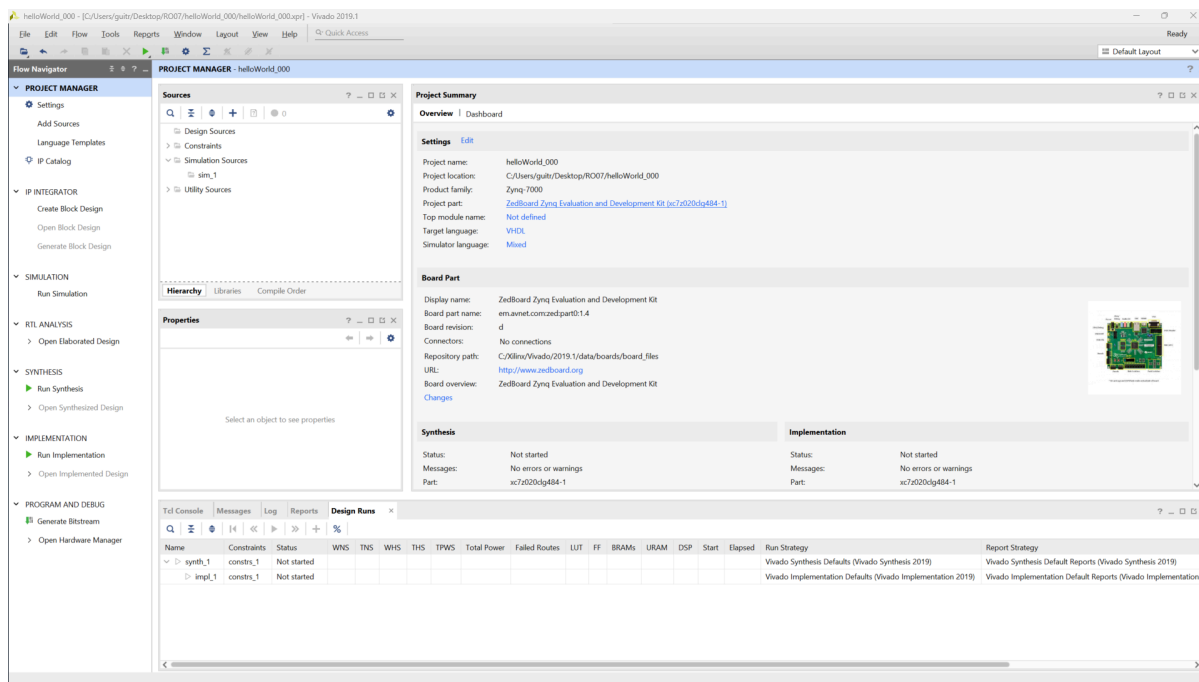
This tutorial is mostly based on this video: [IIITD AELD Lab1\\_P1: Review](#).

## Naming Convention

- `.elf` : programs the ARM core;
- `.bit` : programs the FPGA core;

Vivado configures the hardware part and the SDK configures the software part of the ZedBoard.

## Vivado 2019.1 - Create Project



## Zynq Configuration

## ZYNQ7 Processing System (5.5)

[Documentation](#) [Presets](#) [IP Location](#) [Import XPS Settings](#)

## Page Navigator

Zynq Block Design

PS-PL Configuration

Peripheral I/O Pins

MIO Configuration

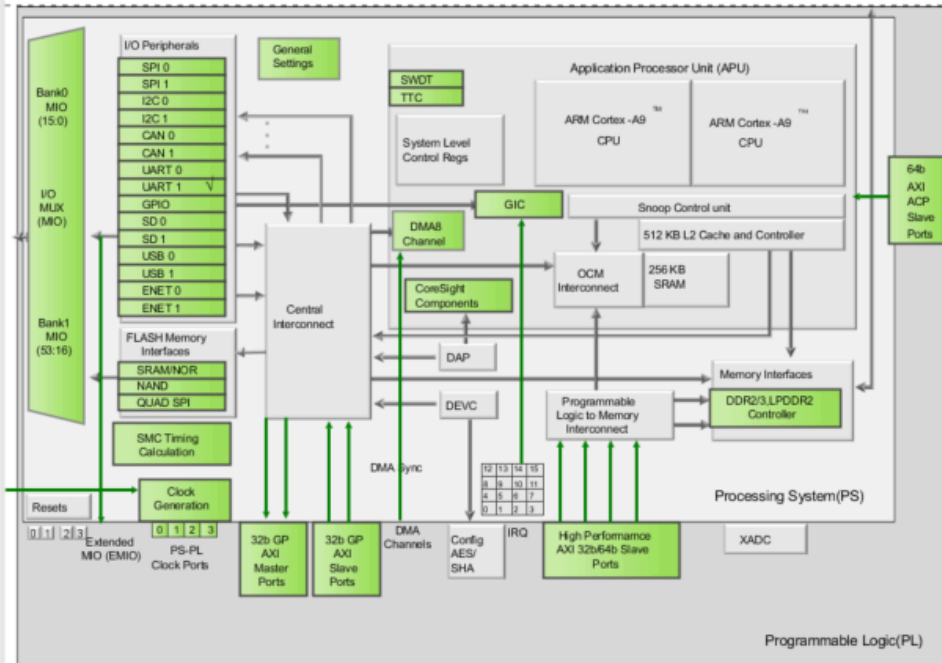
Clock Configuration

DDR Configuration

SMC Timing Calcula

Interrupts

## Zynq Block Design

[Summary Report](#)

OK

Cancel

disable all but Clock Reset 0

Re-customize IP

ZYNQ7 Processing System (5.5)

Documentation

Presets

IP Location

Import XPS Settings

Page Navigator

Zynq Block Design

PS-PL Configuration

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PS-PL Configuration

Summary Report

Search: Q

| Name                      | Select                              | Description                                        |
|---------------------------|-------------------------------------|----------------------------------------------------|
| FLCK_CLKTRIG3             | <input type="checkbox"/>            | Enables PL clock trigger signal 3 used to halt the |
| Enable Clock Resets       |                                     |                                                    |
| FCLK_RESET0_N             | <input checked="" type="checkbox"/> | Enables general purpose reset signal 0 for PL logi |
| FCLK_RESET1_N             | <input type="checkbox"/>            | Enables general purpose reset signal 1 for PL logi |
| FCLK_RESET2_N             | <input type="checkbox"/>            | Enables general purpose reset signal 2 for PL logi |
| FCLK_RESET3_N             | <input type="checkbox"/>            | Enables general purpose reset signal 3 for PL logi |
| AXI Non Secure Enablement | 0                                   | Enable AXI Non Secure Transaction                  |
| GP Master AXI Interface   |                                     |                                                    |
| > M AXI GP0 interface     | <input type="checkbox"/>            | Enables General purpose AXI master interface 0     |
| > M AXI GP1 interface     | <input type="checkbox"/>            | Enables General purpose AXI master interface 1     |
| GP Slave AXI Interface    |                                     |                                                    |
| S AXI GP0 interface       | <input type="checkbox"/>            | Enables General purpose 32-bit AXI Slave interfac  |
| S AXI GP1 interface       | <input type="checkbox"/>            | Enables General purpose 32-bit AXI Slave interfac  |

OK

Cancel

disable all but UART 1



ings

### Peripheral I/O Pins

## Summary Report

Search:

## Peripherals

Bank 0 LVCMOS 3.3V

Bank 1

> ☐ Quad SPI Flash

> ☐ SRAM/NOR Flash

> ☐ NAND Flash

> ☐ Ethernet 0

> ☐ Ethernet 1

☐ USB 0☐ USB 1

> ☐ SD 0

> ☐ SD 1

> ☐ SPI 0

> ☐ SPI 1

> ☐ UART

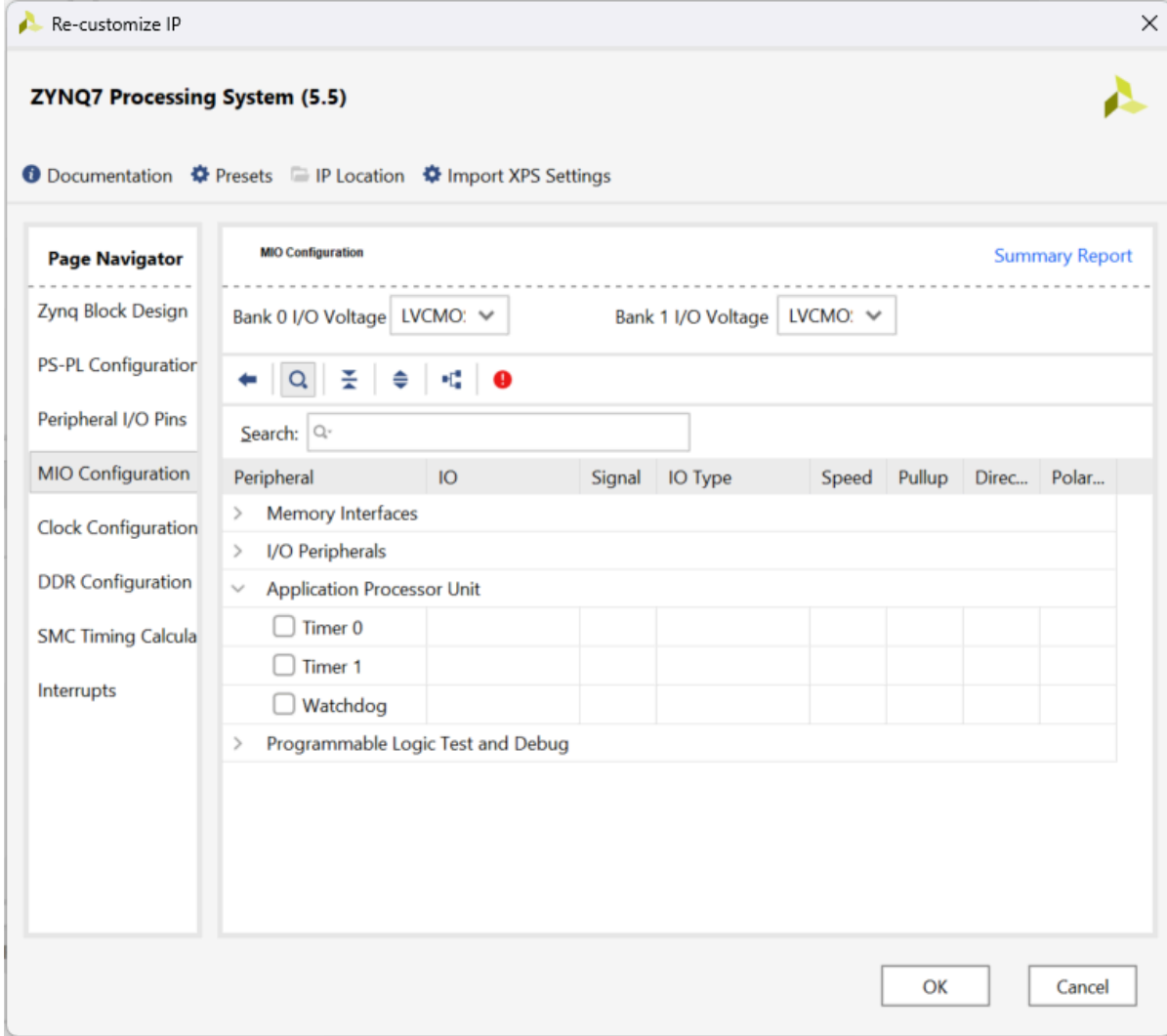
✓ ☒ UART

☐ Modem Signa☐ I2C 0

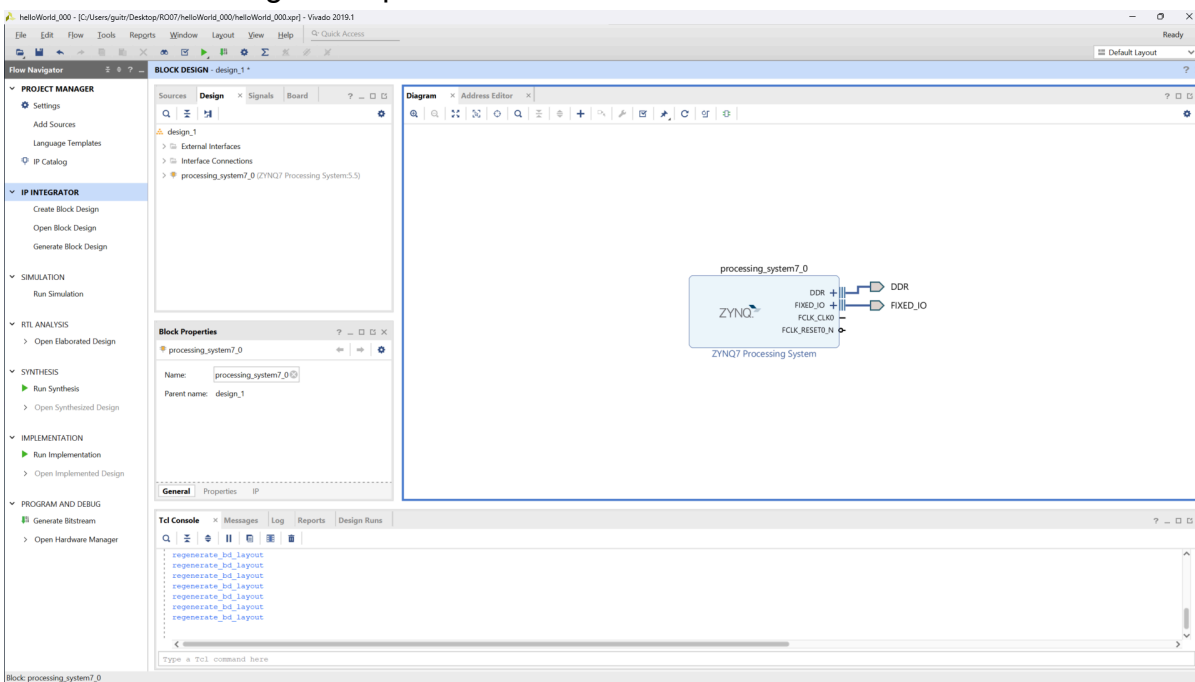
OK

Cancel

disable Timer 0



finished block design after procedure



[Vivado 2019.1 - Create Project](#)

[Vivado 2019.1 - Create Block Diagram](#)

[Vivado 2019.1 - Generate Output Products](#)

[Vivado 2019.1 - Export Block Design](#)

[Vivado 2019.1 - Generate Bitstream](#)

[Vivado 2019.1 - Export Hardware](#)

[Vivado SDK 2019.1 - Application Project](#)

[Vivado SDK 2019.1 - Board Connection](#)

[Vivado SDK 2019.1 - Configuration UART](#)

[Vivado SDK 2019.1 - Debug Configuration](#)

[Vivado SDK 2019.1 - JTAG Terminal](#)

**Execution**

helloWorld\_000.sdk - Debug - helloWorld\_000/src/helloworld.c - Xilinx SDK

File Edit Source Refactor Navigate Search Project Run Xilinx Window Help

Quick Access

Debug Resume (F8)

System Debugger using Debug\_helloWorld\_000.elf on Local (Local)

APU

- ARM Cortex-A9 MPCore #0 (Breakpoint: main)
  - 0x0010058c main(): ./src/helloworld.c, line 55
  - 0x001007f0 \_start(): xil-crt0.S, line 138
  - ...
- ARM Cortex-A9 MPCore #1 (Suspended)

xc7z020

XSCT Process

```
xsct% jtagterminal
64088
xsct% jtagterminal
64974
xsct%
```

xsct%

system.hdf system.mss helloworld.c

```
/* Copyright (C) 2009 - 2014 Xilinx, Inc. All rights reserved.

/*
 * helloworld.c: simple test application
 *
 * This application configures UART 16550 to baud rate 9600.
 * PS7 UART (Zynq) is not initialized by this application, since
 * bootrom/bsp configures it to baud rate 115200
 *
 * -----
 * | UART TYPE   BAUD RATE |
 * -----
 * uarts550     9600
 * uartlite     Configurable only in HW design
 * ps7_uart     115200 (configured by bootrom/bsp)
 */

#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"

int main()
{

```

Outline

- stdio.h
- platform.h
- xil\_printf.h
- main() : int

Console Tasks SDK Terminal Problems Executables

TCF Debug Virtual Terminal - ARM Cortex-A9 MPCore #1

SDK Log Memory

Monitors

Writable Smart Insert 55 : 1

JTAG-based Hyperterminal.

Connected to JTAG-based Hyperterminal over TCP port : 62995

(using socket : sock716)

Help :

Terminal requirements :

(i) Processor's STDOUT is redirected to the ARM DCC/MDM UART

(ii) Processor's STDIN is redirected to the ARM DCC/MDM UART.

Then, text input from this console will be sent to DCC/MDM's UART port.

NOTE: This is a line-buffered console and you have to press "Enter"  
to send a string of characters to DCC/MDM.

Hello World

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